Circuit tools and semiconductor physics

Answers and worked solutions · Day 01 · Sat 24 Oct · pack E1

5. Practice

From your course files

WS1 Q1 (course question).

> **a) Negative.** More electrons than protons means a net negative charge.

WS1 Q2 (course question).

> **d) Right, Left.** Conventional current is the flow of positive charge, so i_1 (arrow pointing right) is positive when positive charges move right, or equivalently when negative charges move left.

WS1 Q3 (course question).

> **c) Negative.** A wrongly guessed polarity just gives a negative answer. The magnitude is still right.

WS1 Q4 (course question).

> **a) Lose, Positive, Negative, Absorbed.** A resistor always absorbs energy and dissipates it as heat, so carriers lose potential energy going from + to −.

WS1 Q5 (course question).

> **b) Always positive** (official answer). If $i < 0$ then $v < 0$ too, so $p = vi = i^2R$. Strictly $p \geq 0$, and it is zero when $i = 0$.

WS1 Q6 (course question).

> **b) Open circuit.** $i = C\,dv/dt = 0$ when v is constant, and zero current is an open circuit.

WS1 Q7 (course question).

> **a) Short circuit.** $v = L\,di/dt = 0$ when i is constant, and zero voltage is a short circuit.

WS1 Q8 (course question).

> **a) Low, High, Positive.** Active sign convention: current flows from the − end to the + end inside the source, and a source that supplies energy has positive $i(t)$.

WS1 Q9 (course question).

> **b) Positive, Positive.** In the active convention a supplying source has $v(t) > 0$, and positive current leaves the + end.

WS1 Q10 (course question).

> From left to right: **VCVS** ($v = k \cdot v_1$), **CCVS** ($v = k \cdot i_1$), **VCCS** ($i = k \cdot v_1$), **CCCS** ($i = k \cdot i_1$). Diamond with ± means a voltage source. Diamond with an arrow means a current source. The controlling variable is v_1 or i_1.

WS1 Q11 (course question).

> **a, c, f.** A short has zero voltage, zero resistance and infinite conductance. Its current is whatever the circuit sets.

WS1 Q12 (course question).

> **a, c.** A charged capacitor or inductor can give energy back. With the passive convention it delivers when $p < 0$ and absorbs when $p > 0$.

WS1 Q13 (course question).

> **c) $N = 3, B = 5$.** The curved wire joins the two top-left and top-right nodes of the 5 kΩ/10 kΩ pair, so those nodes are one node. The 20 kΩ then has both ends on the same node: it is shorted and is not a branch. That leaves three nodes (top, middle, bottom) and five elements: 12 V, 5 kΩ, 10 kΩ (top), 10 kΩ and 5 kΩ (vertical).

WS1 Q14 (course question).

Loop = 6, Mesh = 3 (official). Options a) and d) are printed identically on the sheet, so either letter carries this answer. Count meshes with $B - N + 1$; count loops by listing every closed path that uses no node twice.

WS1 Q15 (course question).

a). The horizontal 5 Ω is open at its right end, so $I_1 = 0$. KCL at the top-left node: $9 = 3 + I_2$, so $I_2 = 6$ A.

WS1 Q16 (course question).

a). Go round the blue loop in its (anticlockwise) direction and take each voltage's sign as you meet it: top resistor $-v_1$, A–B resistor $+v_4$, B–D resistor $-v_3$, D–C resistor $+v_2$. The loop does not pass through the 3 A source, so v_5 is not in it.

WS1 Q17 (course question).

c) 16 V. Everything is in series, so the sources can be collected. Taking the 9 V polarity as positive: $+9 + 3 - 1 + 5 = 16$ V. (The series resistance is $100 + 220 = 320\ \Omega$.)

WS1 Q18 (course question).

b) 4.8 A. The right-hand 2 Ω and 6 Ω are in series (8 Ω), in parallel with the middle 2 Ω. Current divider: $I_1 = 6 \times \dfrac{8}{8+2} = 4.8$ A.

WS1 Q19 (course question).

c) 7.2 V. The middle 2 Ω has $2 \times 4.8 = 9.6$ V across it, and so does the 8 Ω branch. Voltage divider: $V_2 = 9.6 \times \dfrac{6}{2+6} = 7.2$ V.

WS1 Q20 (course question).

c) 0.66 A.

- 2 A on, 3.5 V off (short). The short puts the 5 Ω out of circuit, so 2 A splits between the 7+3 = 10 Ω leg and the 15 Ω: $I_{x1} = 2 \times \dfrac{10}{10+15} = 0.8$ A.
- 3.5 V on, 2 A off (open). The 3.5 V sits across the 15 Ω + 10 Ω path, driving current against the I_x arrow: $I_{x2} = -\dfrac{3.5}{15+7+3} = -0.14$ A.

- Sum: $I_x = 0.8 - 0.14 = 0.66$ A.

WS1 Q21 (course question).

a) 45 V, 3 Ω. Remove the 6 Ω load. R_{Th}: short the 36 V, open the 3 A, leaving only the 3 Ω, so $R_{Th} = 3\ \Omega$. V_{Th}: with A–B open, the 3 A flows round through the 3 Ω, dropping $3 \times 3 = 9$ V. KVL round the outer loop: $V_{Th} = 36 + 9 = 45$ V.

WS1 Q22 (course question).

d) 15 A, 3 Ω. $R_N = R_{Th} = 3\ \Omega$. Short A–B: the 3 A source delivers 3 A into the short, and the 36 V drives $36/3 = 12$ A through the 3 Ω into the short. $I_N = 3 + 12 = 15$ A. Check: $I_N = V_{Th}/R_{Th} = 45/3 = 15$ A.

WS1 Q23 (course question).

b) 0.1923 mA. Transform 5 V + 5 kΩ into a Norton source: $5/5\ \mathrm{k} = 1$ mA in parallel with 5 kΩ. Now 1 mA + 1 mA = 2 mA feeds 5 kΩ ‖ 47 kΩ. Current divider:
$I_x = 2 \times \dfrac{5}{5 + 47} = 0.1923$ mA.

WS1 Q24 (course question).

c) $V_c = 0.2$ V. Full working is in Example 2 above: three KCLs (A, C, D) give $V_A = 0.3556$ V, $V_C = 0.2$ V, $V_D = 0.2444$ V.

WS1 Q25 (course question).

a) 0.3556 V (the options are labelled "V_c" on the sheet but mean V_{AB}). The 1 A source is on the outside of mesh 1, so $I_1 = 1$ A. KVL for meshes 2, 3, 4 (conductances, so each resistor drop is current/G):

$$-(1 - I_2) + \frac{I_2 - I_3}{2} + \frac{I_2 - I_4}{2} = 0, \quad \frac{I_3}{3} - \frac{I_4 - I_3}{2} - \frac{I_2 - I_3}{2} = 0, \quad -\frac{I_2 - I_4}{2} + \frac{I_4 - I_3}{2} + I_4 = 0$$

Simplified: $4I_2 - I_3 - I_4 = 2$, $-3I_2 + 8I_3 - 3I_4 = 0$, $-I_2 - I_3 + 4I_4 = 0$. So $I_2 = 0.6444$, $I_3 = 0.3333$, $I_4 = 0.2444$ A, and $V_{AB} = (I_1 - I_2)/1\ \mathrm{S} = 0.3556$ V. This matches V_A from Q24.

Note: the official solution prints the first simplified equation as "$= 0$". It should be "$= 2$" (multiply the first KVL by 2). Its numerical answers satisfy the "$= 2$" version.

WS2 Q1 (course question).

I **F** (it is an insulator at 0 K). II **T**. III **F** (intrinsic Si is not a conductor). IV **T**. V **T**. VI **T**. VII **F** (doped material is neutral: every free carrier is balanced by a fixed ionised dopant). VIII **T**. IX **T**. X **T**.

WS2 Q2 (course question).

$L = 0.8$ cm, $D = 0.1$ cm, $A = \pi D^2/4 = 7.854 \times 10^{-3}$ cm². $\rho = 1/\sigma = 1.77 \times 10^{-6}$ Ω-cm.

$$R = \frac{\rho L}{A} = \frac{1.77 \times 10^{-6} \times 0.8}{7.854 \times 10^{-3}} = \mathbf{1.80 \times 10^{-4}\ \Omega}$$

WS2 Q3 (course question).

(a) $n = p = n_i$, so
$\sigma = q(\mu_n + \mu_p) n_i = 1.6 \times 10^{-19}(1850)(1.5 \times 10^{10}) = 4.44 \times 10^{-6}\ (\Omega\text{-cm})^{-1}$ and $\rho = 2.252 \times 10^{5}\ \Omega$-cm. $R = \rho L/A = 2.252 \times 10^{5} \times 0.8/7.854 \times 10^{-3} =$ **22.9 MΩ** (the official 22.95 MΩ rounds A to 7.85×10^{-3}).

(b) $E = V/L = 5/0.8 = 6.25$ V/cm. $v_{dn} = \mu_n E = 8.44 \times 10^{3}$ **cm/s**, $v_{dp} = \mu_p E = 3.13 \times 10^{3}$ **cm/s**.

WS2 Q4 (course question).

(a) $N_D \gg n_i$, so $n_n \approx N_D = 10^{14}$ **cm⁻³** and $p_n = n_i^2/N_D = (1.5 \times 10^{10})^2/10^{14} = 2.25 \times 10^{6}$ **cm⁻³**.

(b) The hole term is negligible: $\sigma \approx q N_D \mu_n = 1.6 \times 10^{-19} \times 10^{14} \times 1350 = 0.0216\ (\Omega\text{-cm})^{-1}$.

WS2 Q5 (course question).

Worked in full in Example 4 above. Final answers: 300 K, $p \approx 2 \times 10^{15}$, $n = 1.125 \times 10^{5}$ cm⁻³; 405 K, $n = 5 \times 10^{10}$ cm⁻³. The exact equations change these only in the 5th significant figure.

WS2 Q6 (course question).

Worked in full in Example 3 above. Final answers: (a) $n_i = 7.88 \times 10^{10}$ cm⁻³; (b) $n = p = n_i$; (c) $\rho = 4.29 \times 10^{4}\ \Omega$-cm; (d) $R = 238$ MΩ; (e) 81 % lower than the 1.25 GΩ

at 300 K.

WS2 Q7 (course question).

Worked in full in Example 3, step 5. $\rho = 0.926\ \Omega$-cm and $R = 5.14$ kΩ. The official solution refers to "Problem 5" and works at 300 K. At 50 °C the minority term is still negligible ($p_n = 1.24 \times 10^6$), so $R = 5.14$ kΩ either way, assuming the mobilities do not change.

PT Q1 (practice test, 5 marks).

(I) Answer: a. Electrons and holes recombine near the junction and leave behind ionised dopants. (b) is wrong because the bulk still conducts. (c) is wrong because only carriers near the junction cross, and equilibrium stops further flow. (d) is wrong because a depletion region exists with no applied voltage at all.

(II) Answer: a. $r_d = \Delta V_d / \Delta I_d$ is the inverse slope of the I–V curve at the Q-point. This is covered in pack E2.

(III) Answer: b. Full-wave rectification uses both half-cycles.

(IV) Answer: a. A thin, lightly doped base means few injected electrons recombine, so most reach the collector (WS4 §3.2).

(V) Answer: b. The DC analysis fixes the Q-point. The AC analysis studies the small variations about it.

New questions

N1. [5 marks] Circle the correct answer (I to V).

I – b. $p = n_i^2 / N_D = (1.5 \times 10^{10})^2 / 10^{16} = 2.25 \times 10^4\ \text{cm}^{-3}$. Option (d) is n_i^2 itself, and (a) is n_i, which applies only to intrinsic material.

II – b. At equilibrium, the majority-carrier diffusion current I_D is balanced by the minority-carrier drift current I_S. Carriers do cross the junction, so (a) is wrong. The depletion region holds only bound ions, so (d) is wrong.

III – c. Charge balance gives $x_n N_D = x_p N_A$, so $x_n / x_p = N_A / N_D = 100$. The region extends into the lightly doped n side.

IV – b. Short V_{CC}. Both resistors then run from the base to ground, in parallel.

V – a. These are the two ideal-op-amp rules (WS4 §2).

N2. [6 marks]

(a) Since $N_A \gg n_i$: $p \approx N_A = 4 \times 10^{15}\ \text{cm}^{-3}$ and $n = n_i^2/N_A = (1.5 \times 10^{10})^2/(4 \times 10^{15}) = 5.625 \times 10^{4}\ \text{cm}^{-3}$.

(b) $\sigma = q(n\mu_n + p\mu_p) = 1.6 \times 10^{-19}\left(5.625 \times 10^{4} \cdot 1350 + 4 \times 10^{15} \cdot 500\right)$. The electron term is 3.8×10^{-11} times the hole term, so it is negligible. $\sigma = 0.320\ (\Omega\text{-cm})^{-1}$ and $\rho = 3.125\ \Omega$**-cm**.

(c) $L = 0.2$ cm and $A = 0.05 \times 0.02 = 1.0 \times 10^{-3}$ cm². $R = \rho L/A = 3.125 \times 0.2/10^{-3} =$ **625 Ω**, so $I = 3/625 =$ **4.8 mA**.

(d) $E = V/L = 3/0.2 = 15$ V/cm, so $v_{dp} = \mu_p E = 500 \times 15 = 7.5 \times 10^{3}$ **cm/s**. (Check: $J = qpv_{dp} = 4.8\ \text{A/cm}^2$, and $J \times A = 4.8$ mA, which agrees.)

N3. [7 marks] (chained)

(a) $V_{Th} = 15 \cdot \frac{12}{68+12} =$ **2.25 V**. $R_{Th} = 68 \parallel 12 = \frac{68 \times 12}{80} =$ **10.2 kΩ**.

(b) KVL around $V_{Th} \to R_{Th} \to 0.7\ \text{V} \to 101$ kΩ:

$$I_B = \frac{V_{Th} - 0.7}{R_{Th} + 101\ \text{k}} = \frac{1.55}{111.2\ \text{k}} = \mathbf{13.94\ \mu A}$$

$V_B = V_{Th} - I_B R_{Th} = 2.25 - 13.94\,\mu\text{A} \times 10.2\ \text{k} =$ **2.108 V**.

(c) Ignoring R_{Th}: $I_B = 1.55/101\ \text{k} = 15.35\ \mu$A. The error is $(15.35 - 13.94)/13.94 =$ **+10.1 %**. This is why the course asks for the Thevenin step (practice test Q7(a)).

N4. [7 marks] (chained)

(a) Short both supplies: $R_{Th} = 20 \parallel 30 \parallel 15$. Then $1/R_{Th} = 1/20 + 1/30 + 1/15 = 0.15\ \text{mS}$, so $R_{Th} = 6.667$ **kΩ**.

(b) KCL at X, open circuit: $\frac{12 - V_X}{20} + \frac{-6 - V_X}{30} + \frac{0 - V_X}{15} = 0$.

$$V_{Th} = \frac{12/20 - 6/30}{0.15} = \frac{0.6 - 0.2}{0.15} = \mathbf{2.667\ V}$$

Superposition:

- +12 V alone: $30 \parallel 15 = 10$ k, so the contribution is $12 \cdot \frac{10}{20+10} = 4.0$ V.
- −6 V alone: $20 \parallel 15 = 8.571$ k, so the contribution is $-6 \cdot \frac{8.571}{30+8.571} = -1.333$ V.
- Sum: 2.667 V. This agrees.

(c) $V_X = V_{Th}\frac{5}{5+6.667} =$ **1.143 V**, so $I_L = 1.143/5\ \text{k} =$ **0.229 mA**. (Check by direct nodal analysis with the 5 kΩ added: the same 1.143 V.)

(d) $I_N = V_{Th}/R_{Th} = 2.667/6.667\ \text{k} = \mathbf{0.400\ mA}$ (flowing into X when X is shorted to ground), in parallel with $R_N = 6.667$ kΩ.

N5. [6 marks]

(a) Currents leaving the node are positive.

$$\text{Node 1: } \frac{V_1 - 12}{2} + \frac{V_1}{4} + \frac{V_1 - V_2}{3} = 0$$
$$\text{Node 2: } \frac{V_2 - V_1}{3} + \frac{V_2}{6} = 1\ \text{mA}$$

(resistances in kΩ, currents in mA). In matrix form:

$$\begin{bmatrix} 1.0833 & -0.3333 \\ -0.3333 & 0.5 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 1 \end{bmatrix}$$

$V_1 = 7.742$ **V and** $V_2 = 7.161$ **V.** Check at node 2: $(7.742 - 7.161)/3 = 0.194$ mA plus 1 mA $= 1.194$ mA, and $7.161/6 = 1.194$ mA. KCL balances.

(b) Current leaves the + terminal of the source (active convention), so $P = V_2 \times 1\ \text{mA} =$ **7.16 mW, supplied**.

(c) Switch off both sources: $R_{Th} = \big((2 \parallel 4) + 3\big) \parallel 6 = (1.333 + 3) \parallel 6 = \mathbf{2.516\ k\Omega}$. (Check: $V_{oc} = V_2 = 7.161$ V and $I_{sc} = 2.846$ mA by direct calculation, so $7.161/2.846 = 2.516$ kΩ.)

N6. [8 marks] (hard, chained)

(a) $n \approx N_D = 10^{13}$ and $p = (1.5 \times 10^{10})^2/10^{13} = 2.25 \times 10^7$.
$\sigma = 1.6 \times 10^{-19}(10^{13} \cdot 1350) = 2.16 \times 10^{-3}\ (\Omega\text{-cm})^{-1}$, so ρ = 463 Ω-cm.

(b) $n_i = 7.3 \times 10^{15}(450)^{3/2} e^{-1.12/(2\times 8.62\times 10^{-5}\times 450)} = 3.744 \times 10^{13}\ \text{cm}^{-3}$. This is larger than N_D.

(c) The approximation gives $p = (3.744 \times 10^{13})^2/10^{13} = 1.40 \times 10^{14}$, which is bigger than the "majority" $n = 10^{13}$. A minority concentration cannot exceed the majority, so the approximation (which assumes $N_D \gg n_i$) fails.

(d) Exact:

$$n = \frac{N_D}{2} + \sqrt{\left(\frac{N_D}{2}\right)^2 + n_i^2} = 5 \times 10^{12} + \sqrt{(5 \times 10^{12})^2 + (3.744 \times 10^{13})^2} = \mathbf{4.277 \times 10^{13}}$$

$p = n_i^2/n = 3.277 \times 10^{13}$. Check: $n - p = 10^{13} = N_D$, so charge neutrality holds. The sample is now close to intrinsic.

(e) $\sigma = 1.6 \times 10^{-19}(4.277 \times 10^{13} \cdot 1350 + 3.277 \times 10^{13} \cdot 500) = 1.186 \times 10^{-2}\ (\Omega\text{-cm})^{-1}$. The ratio to 300 K is **5.49**. $V_T = \frac{1.38 \times 10^{-23} \times 450}{1.6 \times 10^{-19}} = \mathbf{38.8\ mV}$. (In reality mobility falls with temperature, so the true ratio would be smaller. That is why we stated the assumption.)

6. Quick recall check

Question

1. $V_T = kT/q = 1.38 \times 10^{-23} \times 300 / 1.6 \times 10^{-19} = 25.875$ mV ≈ 26 mV.
2. $n_i^2 = 2.25 \times 10^{20}\ \text{cm}^{-6}$, so $p = 2.25 \times 10^{5}\ \text{cm}^{-3}$.
3. A voltage source becomes a short. A current source becomes an open. Dependent sources stay in.
4. The lightly doped side: $x_n / x_p = N_A / N_D$.
5. $V_{Th} = V_{CC} R_2 / (R_1 + R_2)$ and $R_{Th} = R_1 R_2 / (R_1 + R_2)$.